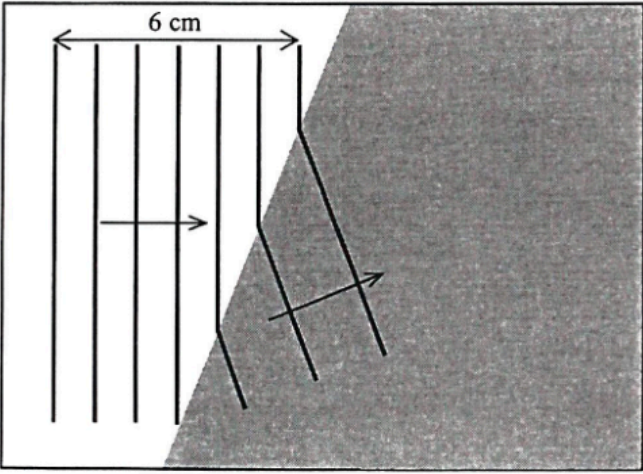


1.	(a)	(i)	$(1.5)(4200)(60 - 10) = m(3.34 \times 10^5) + m(4200)(10 - 0)$ $m = 0.837766 \text{ kg} \approx 0.838 \text{ kg}$	1M+1M 1A	Note : Convection is the least effective transfer process in this case as the hot air outside is above the denser cool air around the ice cream.	
				3		
		(ii)	More ice is needed to cool the container as it will release heat / thermal energy as well.	1A 1A		
			2			
	(b)	(i)	Conduction:	1A		Any ONE
			- Foam is a poor conductor (of thermal energy) and it minimizes the transfer of heat / thermal energy from the surroundings to the cool contents / ice cream (inside the bag).			
			<u>OR</u>			
			Radiation:			
			- The shiny surface (inside) reduces the transfer of heat / thermal energy from the surroundings to the cool contents / ice cream (inside the bag) through emission of radiation			
			<u>OR</u>			
		Convection:				
		- The zipper prevents convection between the hot air outside and the cool contents / ice cream (inside the bag).				
			1			
	(ii)	(Radiation)	Make the outer surface (of the bag) shiny.	1A		
				1		

2. (a)	$pV = nRT$ $(100 \times 10^3)(0.52) = n(8.31)(273+15)$ $n = 21.727504 \text{ (mol)} \approx 21.7 \text{ (mol)}$	1M 1A	Accept: (21.0 ~ 22.0) mol
		2	
(b) (i)	Since $pV = nRT \Rightarrow V = \frac{nRT}{p}$ / volume V of the balloon depends on both T and p, the (fractional) decrease in pressure p (with height) is greater / faster than the (fractional) decrease in temperature T.	1A 1A	Accept: volume V decreases as temperature T decreases would be true only if pressure p is constant, however, p decreases with height.
		2	
(ii) (1)	$\frac{pV}{T} = \text{constant}$ $\frac{(100)(0.52)}{(273+15)} = \frac{p(8)}{216}$ $p = 4.875 \text{ kPa or } 4875 \text{ Pa}$	1M 1A	
		2	
(2)	$p = p_0 e^{-kx}$ $4.875 = 100 e^{-0.138 x}$ $x = 21.89166726 \text{ (km)} \approx 21.9 \text{ (km)}$	1M 1A	Accept: (21.7 ~ 22.0) km
		2	

3. (a) (i) (1) $\frac{1}{2}mv^2 = mgh$ $v^2 = 2(9.81)(12)$ $v = 15.344054 \text{ m s}^{-1} \approx 15.3 \text{ m s}^{-1}$ $(v = 15.491933 \text{ m s}^{-1} \approx 15.5 \text{ m s}^{-1} \text{ for } g = 10 \text{ m s}^{-2})$	1M	Accept: (15.0 ~ 15.5) m s ⁻¹
	1A	
	2	
(2) $s = \frac{1}{2}gt^2$ $12 = \frac{1}{2}(9.81)t^2$ $t = 1.564124 \text{ s} \approx 1.56 \text{ s}$ $(t = 1.5491933 \text{ s} \approx 1.55 \text{ s} \text{ for } g = 10 \text{ m s}^{-2})$	1M	Accept: (1.50 ~ 1.60) s
	1A	
	2	
(ii) $F - mg = \frac{mv - mu}{t}$ $F = \frac{70 \times (0 - (-15.3))}{0.3} + 70 \times 9.81$ $= 4266.9793 \text{ N} \approx 4270 \text{ N}$ $(F = 4314.7845 \text{ N} \approx 4310 \text{ N} \text{ for } g = 10 \text{ m s}^{-2})$	1M+1M	Accept: (4180 ~ 4320) N
	1A	
	3	
(iii) Elastic potential energy	1A	1
(b) (i) (Velocity is too high, hence the force for deceleration is very large.) - The life net may be torn. - The falling person may be injured. - The firemen may not be able to hold the life net tight.	1A	Any ONE
	1	
(ii) There exists a horizontal velocity when a person jumps and the horizontal displacement is very difficult to estimate as it depends on the time of fall, which is relatively long.	1A	1A
	2	

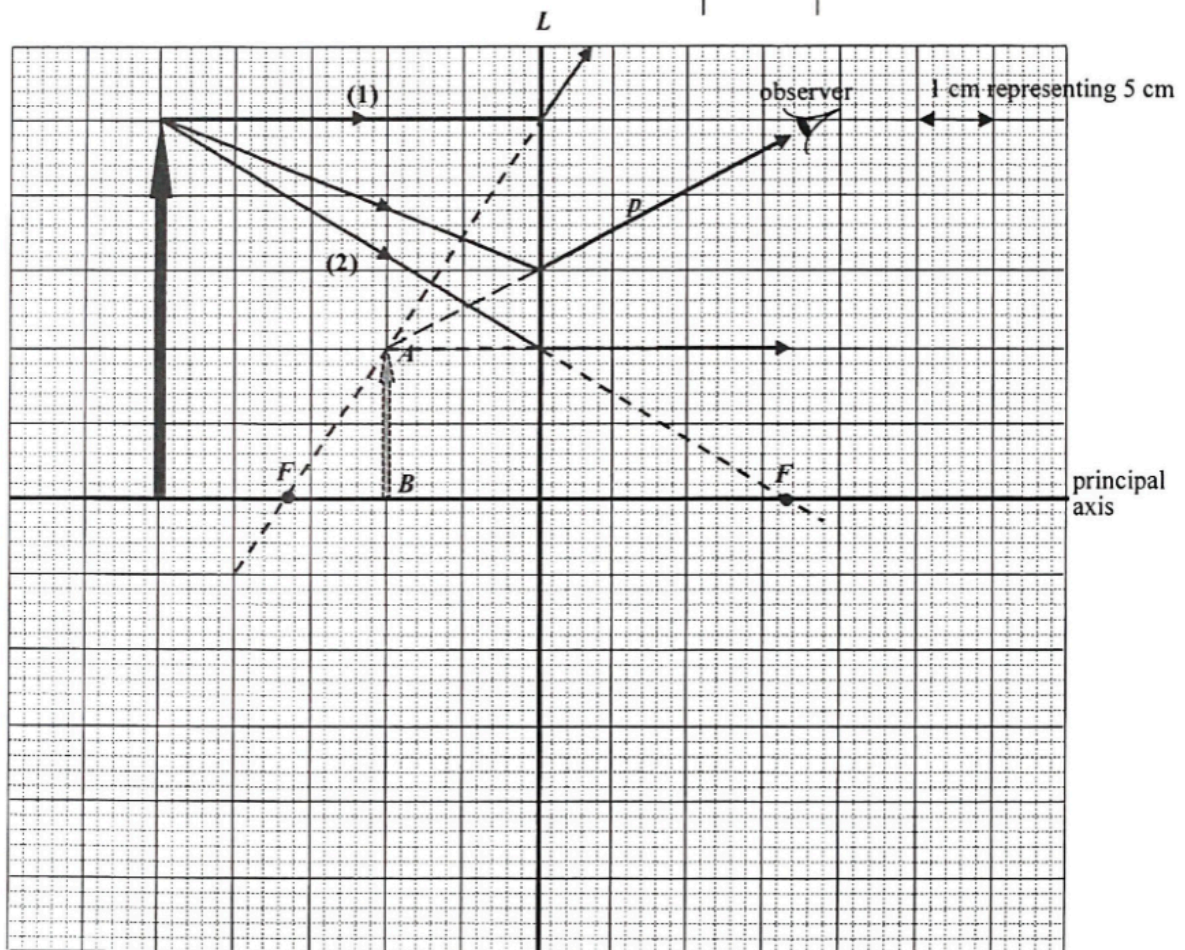
5. (a) (i) wavelength $\lambda = \frac{0.06}{7-1}$ $= 0.01 \text{ m } (= 1 \text{ cm})$	1A 1	
(ii) speed $v = f\lambda = 10 \times 0.01$ $= 0.1 \text{ m s}^{-1} (= 10 \text{ cm s}^{-1})$	1M/1A 1	
(b) (i) frequency = 10 Hz	1A 1	
(ii) 	1A 1A 2	
(iii) Refraction. It is due to the change in wavelengths / wave speeds in different media / depths.	1A 1A 2	

6. (a) L is diverging / concave.
Only diverging / concave lens forms diminished, virtual image.

1A
1A

2

(b)



Correct position and height of object

2A

2

- (c) Correct ray to locate F and focus F correctly marked.
Focal length = 16.5 cm

2M

1A

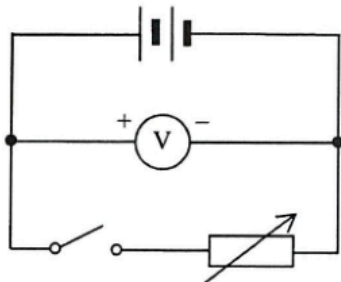
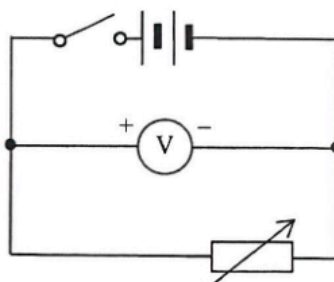
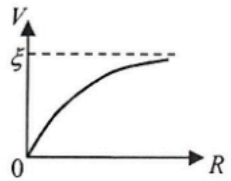
Accept: (15.5 ~ 17.5) cm

3

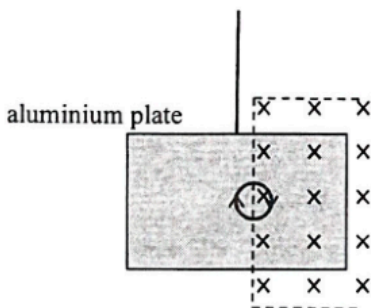
- (d) Correct ray p from tip of object

2A

2

7. (a)  Close the switch and record corresponding V and R readings Adjust the resistance R to lower/other value(s) and repeat the experiment Precaution: - First set the variable resistor to its maximum / a large value - Open the switch after each measurement - Any reasonable answer	1A 1A	Correct circuit with correct symbol Correct polarity Alternative circuit 
(b) Terminal voltage V delivered increases with increasing (loading) resistance R (or graphical representation) $V = \xi \frac{R}{R + r} \quad \text{OR} \quad V = \xi - \frac{\xi}{R + r} r$	1A 1A	Accept: 

8. (a)		1A	1
(b) (i) - If one of the lighting sets / circuits fails, the other (in parallel) can still operate, i.e. both work independently. - Both can work at the rated power. - Any reasonable answer	Any ONE	1A	1
(ii) $P = IV$ $(300 + 450) = I(220)$ $I = 3.409091 \text{ A} \approx 3.41 \text{ A}$		1M 1A	3
Thus 5 A fuse should be used.		1A	
(c) Electrical energy used per day $= 0.500 \text{ kW} \times 8 \text{ h} + 2 \text{ kW} \times 0.5 \text{ h} + 3 \text{ kW} \times 2 \text{ h}$ $= 11 \text{ kW h}$ Cost $= \$0.9 / \text{kW h} \times 11 \text{ kW h}$ $= \$9.9$		1M 1A	3

9. (a) (i)	By Lenz's law, an e.m.f. would be induced such that it opposes the change, i.e. decrease of magnetic flux (into the paper) by driving an induced current (clockwise) in the coil / circuit (complete).	1A	
		1A	
		2	
		1A	
		1M	
		1A	
		3	
(b) (i)	The change of (magnetic) flux linkage is double that in (a)(ii), i.e. 0.06 Wb.	1M/1A	
		1	
(ii)	Direction of current : $PQRS$	1A	
		1	
(c) (i)		1A	
		1A	
(ii)	Move / swing to the right initially / momentarily / briefly.	1A	
		1	

Solution	Marks	Remarks
10. (a) (i) $x = 3$	1A	
	1	
(ii) More neutrons are produced in each fission for triggering further fissions, i.e. $x > 1$.	1A	
	1	
(b) (i) $m = m_0 e^{-kt}$ $k = \frac{\ln 2}{t_{1/2}} (= 9.846 \times 10^{-10} \text{ yr}^{-1})$ $0.06 = m_0 e^{-\ln 2 \times \left[\frac{2 \times 10^9}{7.04 \times 10^8} \right]}$ $m_0 = 0.429882832 \text{ (kg)} \approx 0.430 \text{ (kg)}$	1M 1A 2	
(ii) $\frac{0.430}{13.556 + 0.430} = 0.03073691 \approx 3.1 \% > 3\%$ Thus natural nuclear fission was possible.	1M/1A 1	
(c) Underground water might run dry.	1A	
<u>OR</u> Energy released by fission dries up the underground water.	1A	
Therefore, fission might stop without slow neutrons.	1A	
	2	

OR $0.06 = m_0 \left(\frac{1}{2} \right)^{\frac{2 \times 10^9}{7.04 \times 10^8}}$